

Homework Assignment 2

Solution Manual

Solutions to all seven questions. Equivalent correct arguments are acceptable. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$.

Question 1 Set notation and set identities

(a) Membership and inclusion.

The three elements of A are $\emptyset$, $\{1\}$, and $\{1, 2\}$.

- (i) $\emptyset \in A$ is **true**: the empty set is explicitly listed as an element.
- (ii) $\emptyset \subseteq A$ is **true**: the empty set is a subset of every set. It has no element that could fail to belong to A .
- (iii) $\{\emptyset\} \subseteq A$ is **true**: its only element, $\emptyset$, belongs to A .
- (iv) $\{1\} \subseteq A$ is **false**: it would require $1 \in A$, but the integer 1 is not one of the three listed elements.
- (v) $\{1\} \in A$ is **true**: the set $\{1\}$ is a listed element.
- (vi) $\{\{1, 2\}\} \in \mathcal{P}(A)$ is **true**: its only element is $\{1, 2\}$, which belongs to A . Thus $\{\{1, 2\}\} \subseteq A$, so it belongs to $\mathcal{P}(A)$.

(b) Cardinalities.

There are three distinct elements in A , so $|A| = 3$ and $|\mathcal{P}(A)| = 2^3 = 8$. An element of A may itself be a set; we count each such set as one element, rather than counting the objects inside it. In particular, 1 , $\{1\}$, and $\{\{1\}\}$ are different objects.

(c) The set identity.

For an arbitrary element u ,

$$\begin{aligned}
 u \in X \setminus (Y \cup Z) &\iff u \in X \wedge \neg(u \in Y \vee u \in Z) \\
 &\iff u \in X \wedge u \notin Y \wedge u \notin Z \\
 &\iff (u \in X \wedge u \notin Y) \wedge (u \in X \wedge u \notin Z) \\
 &\iff u \in (X \setminus Y) \cap (X \setminus Z).
 \end{aligned}$$

The sets have exactly the same elements, so they are equal. The second equivalence is De Morgan's law; the third repeats a condition without changing its truth value.

Question 2 Counting disjoint pairs of subsets

Let $P(n)$ assert that every n -element set has exactly 3^n ordered pairs of disjoint subsets.

Base case ($n = 0$). The only subset of the empty set is $\emptyset$, so the only pair is $(\emptyset, \emptyset)$. It is a disjoint pair, and the count is $1 = 3^0$.

Inductive step. Assume $P(n)$ for an integer $n \geq 0$. Let S be any set of size $n + 1$, choose $x \in S$, and put $T = S \setminus \{x\}$. By the induction hypothesis, T has 3^n ordered pairs (U, V) of disjoint subsets. Each gives three pairs of disjoint subsets of S :

$$(U, V), \quad (U \cup \{x\}, V), \quad (U, V \cup \{x\}).$$

These correspond to x belonging to neither subset, only the first, or only the second. It cannot belong to both because the subsets must be disjoint. Thus the three cases are mutually exclusive and exhaust all possibilities. Removing x from any resulting pair recovers a unique pair (U, V) , so each case contains exactly 3^n pairs, with no repetitions. The total is

$$3^n + 3^n + 3^n = 3 \cdot 3^n = 3^{n+1}.$$

This proves $P(n + 1)$ and completes the induction.

Question 3 Countably infinite sets

(a) Pairing natural numbers with multiples of 3.

Define $f : \mathbb{N} \rightarrow M$ by $f(n) = 3n$. The first six pairings are

n	0	1	2	3	4	5
$f(n)$	0	3	6	9	12	15

Every element $m \in M$ is of the form $m = 3k$ for some $k \in \mathbb{N}$, so it is paired with k . No two different natural numbers are paired with the same element: $3k = 3\ell$ implies $k = \ell$. Thus the pairing reaches every element of M exactly once. Its inverse sends $m \in M$ to $m/3$, which is a natural number. Consequently, M is countably infinite.

(b) Listing the integers that are not multiples of 3.

The first twelve elements are

$$1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17.$$

For each $k = 0, 1, 2, \dots$, list $3k + 1$ followed by $3k + 2$. The list is strictly increasing: within each pair, $3k + 1 < 3k + 2$, and the next pair begins with $3(k + 1) + 1 = 3k + 4 > 3k + 2$. Hence it has no repetitions.

Every element of R has exactly one of the forms $3k + 1$ or $3k + 2$ for a natural number k . Starting with index 0, the number $3k + 1$ appears at index $2k$, and the number $3k + 2$ appears at index $2k + 1$. Both indices are finite, so no element is missed. The listing continues for every $k \in \mathbb{N}$, and all its values are distinct. Therefore R is countably infinite.

For completeness, the listing can be written as the explicit map $h : \mathbb{N} \rightarrow R$ given by

$$h(2k) = 3k + 1, \quad h(2k + 1) = 3k + 2 \quad (k \in \mathbb{N}).$$

Every natural-number index is either $2k$ or $2k + 1$ for exactly one $k \in \mathbb{N}$, so this defines the entire listing.

(c) A proper subset with the same cardinality.

The argument is not valid. Although $M \subsetneq \mathbb{N}$ (for example, $1 \notin M$), part (a) gives a one-to-one pairing between all of $\mathbb{N}$ and all of M . They therefore have the same cardinality.

For a finite set, a proper subset has strictly smaller cardinality. That conclusion does not hold for every infinite set: the pairing $n \leftrightarrow 3n$ gives a concrete counterexample.

Question 4 Ordinary induction**(a) The sum of the first n odd integers.**

Let $P(n)$ be $\sum_{k=1}^n (2k-1) = n^2$. For $n = 0$, the empty sum is $0 = 0^2$. Assume $P(n)$ for some integer $n \geq 0$. Then

$$\begin{aligned} \sum_{k=1}^{n+1} (2k-1) &= \sum_{k=1}^n (2k-1) + (2(n+1)-1) \\ &= n^2 + 2n + 1 && \text{by the induction hypothesis} \\ &= (n+1)^2. \end{aligned}$$

This proves $P(n+1)$, hence the identity holds for every $n \geq 0$.

(b) Divisibility by 8.

Let $P(n)$ be $8 \mid (3^{2n} - 1)$. For $n = 0$, $3^0 - 1 = 0 = 8 \cdot 0$. Assume $P(n)$ for an integer $n \geq 0$, so $3^{2n} - 1 = 8q$ for some integer q . Then

$$\begin{aligned} 3^{2(n+1)} - 1 &= 9 \cdot 3^{2n} - 1 \\ &= 9(3^{2n} - 1) + 8 \\ &= 9 \cdot 8q + 8 = 8(9q + 1). \end{aligned}$$

Since $9q + 1$ is an integer, $P(n+1)$ holds. Induction proves the claim.

(c) A geometric sum.

Let $P(n)$ be $\sum_{k=0}^n 2^k = 2^{n+1} - 1$. For $n = 0$, both sides equal 1. Assume $P(n)$ for an integer $n \geq 0$. Then

$$\begin{aligned} \sum_{k=0}^{n+1} 2^k &= \sum_{k=0}^n 2^k + 2^{n+1} \\ &= (2^{n+1} - 1) + 2^{n+1} \\ &= 2^{n+2} - 1. \end{aligned}$$

This is $P(n+1)$, so the identity holds for every $n \geq 0$.

Question 4 continued**(d) An exponential inequality.**

Let $P(n)$ be $2^n \geq n^2$, for integers $n \geq 4$.

Base case ($n = 4$). We have $2^4 = 16 = 4^2$.

Inductive step. Assume $P(n)$ for an integer $n \geq 4$. Multiplying the induction hypothesis by 2 gives

$$2^{n+1} = 2 \cdot 2^n \geq 2n^2.$$

Also, because $n - 1 \geq 3$,

$$2n^2 - (n + 1)^2 = n^2 - 2n - 1 = (n - 1)^2 - 2 \geq 9 - 2 = 7 > 0.$$

Thus $2^{n+1} \geq 2n^2 \geq (n + 1)^2$, which proves $P(n + 1)$. Induction proves the inequality for every $n \geq 4$.

(e) A recursively defined sequence.

Let $P(n)$ be $a_n = 2 \cdot 3^n - 1$.

Base case ($n = 0$). The defining value is $a_0 = 1$, and $2 \cdot 3^0 - 1 = 1$.

Inductive step. Assume $P(n)$ for an integer $n \geq 0$. Using the recurrence and then the induction hypothesis,

$$\begin{aligned} a_{n+1} &= 3a_n + 2 \\ &= 3(2 \cdot 3^n - 1) + 2 \\ &= 2 \cdot 3^{n+1} - 3 + 2 \\ &= 2 \cdot 3^{n+1} - 1. \end{aligned}$$

Thus $P(n + 1)$ holds, and the formula is valid for every $n \geq 0$.

Question 5 Counting with inclusion–exclusion**(a) Exactly one language.**

Subtract the two overlaps from each individual set size, then add back the triple overlap because it was subtracted twice:

$$\begin{aligned}\text{Python only} &= 110 - 45 - 35 + 15 = 45, \\ \text{Java only} &= 85 - 45 - 30 + 15 = 25, \\ \text{C++ only} &= 70 - 35 - 30 + 15 = 20.\end{aligned}$$

These groups are disjoint, so exactly $45 + 25 + 20 = \boxed{90}$ students have used exactly one language.

(b) Exactly two languages.

Remove the triple overlap from each pairwise intersection:

$$\begin{aligned}\text{Python and Java only} &= 45 - 15 = 30, \\ \text{Python and C++ only} &= 35 - 15 = 20, \\ \text{Java and C++ only} &= 30 - 15 = 15.\end{aligned}$$

The three groups are disjoint, so exactly $30 + 20 + 15 = \boxed{65}$ students have used exactly two languages.

(c) At least one language, and none.

Inclusion–exclusion gives

$$\begin{aligned}|P \cup J \cup C| &= |P| + |J| + |C| - |P \cap J| - |P \cap C| - |J \cap C| \\ &\quad + |P \cap J \cap C| \\ &= 110 + 85 + 70 - 45 - 35 - 30 + 15 \\ &= \boxed{170}.\end{aligned}$$

This also equals $90 + 65 + 15$: those who used exactly one, exactly two, or all three languages. Of the 200 students surveyed, $200 - 170 = \boxed{30}$ have used none of these languages.

Question 6 Understanding a halving algorithm**(a) Trace for $n = 20$.**

Stage	x	count
Initially	20	0
After iteration 1	10	1
After iteration 2	5	2
After iteration 3	2	3
After iteration 4	1	4

In the third iteration, rounding down gives $\lfloor 5/2 \rfloor = 2$. The loop stops when $x = 1$, because the condition $x > 1$ is false. The returned value is $\boxed{4}$.

(b) Powers of two.

Input n	Returned value
1	0
8	3
16	4
32	5

For $n = 2^k$, each iteration reduces the exponent by one:

$$2^k \longrightarrow 2^{k-1} \longrightarrow \dots \longrightarrow 2^0 = 1.$$

There are exactly k halvings, so the returned value is $\boxed{k}$. When $k = 0$, the input is already 1 and the loop runs zero times. Rounding does not affect these calculations, because every power of two greater than 1 is even.

Question 6 continued**(c) Two recurrences and their growth.**

For $n = 2^k$ with $k \geq 0$, the exact formulas are

$$\boxed{T_1(n) = \log_2 n}, \quad \boxed{T_2(n) = n - 1}.$$

We prove the equivalent statements $T_1(2^k) = k$ and $T_2(2^k) = 2^k - 1$ by induction on k . Halving n reduces its exponent by one, so this induction follows the recurrences.

First recurrence: $T_1(n) = T_1(n/2) + 1$.

Base case ($k = 0$). The prescribed initial value is $T_1(2^0) = T_1(1) = 0$, as claimed.

Inductive step. Assume $T_1(2^k) = k$ for an integer $k \geq 0$. Then the recurrence gives

$$T_1(2^{k+1}) = T_1(2^k) + 1 = k + 1.$$

This proves the claim for $k + 1$. Therefore $T_1(2^k) = k$ for every $k \geq 0$, or equivalently $T_1(n) = \log_2 n$ on the stated domain.

Second recurrence: $T_2(n) = 2T_2(n/2) + 1$.

Base case ($k = 0$). We have $T_2(2^0) = T_2(1) = 0 = 2^0 - 1$.

Inductive step. Assume $T_2(2^k) = 2^k - 1$ for an integer $k \geq 0$. Then

$$\begin{aligned} T_2(2^{k+1}) &= 2T_2(2^k) + 1 \\ &= 2(2^k - 1) + 1 \\ &= 2^{k+1} - 1. \end{aligned}$$

Thus the claim holds for $k + 1$. Induction proves $T_2(n) = n - 1$ for every positive power-of-two input, including $n = 1$.

Comparison. As n grows through powers of two, $T_1(n)$ grows logarithmically, while $T_2(n)$ grows linearly. The first recurrence counts one smaller computation at each stage; the second counts two. For example, at $n = 16$ their values are 4 and 15.

Question 7 Proving selection sort correct by induction

(a) Trace.

The initial array is $[5, 2, 4, 2, 1]$. The strict comparison in the inner loop retains the first occurrence of a minimum when values tie.

i	Chosen m	Array after the swap
1	5	$[1, 2, 4, 2, 5]$
2	2	$[1, 2, 4, 2, 5]$
3	4	$[1, 2, 2, 4, 5]$
4	4	$[1, 2, 2, 4, 5]$

Iterations 2 and 4 swap a position with itself, so they leave the array unchanged. The returned array is $[1, 2, 2, 4, 5]$.

(b) An induction claim.

Fix an input array of length $n \geq 1$. Let $P(k)$ assert that after k completed outer-loop iterations:

1. The first k entries are in nondecreasing order.
2. Every entry in the first k positions is at most every entry in positions $k + 1, \dots, n$.
3. For every integer v , the number of occurrences of v equals its number of occurrences in the original input.

The third condition says that the current array is a rearrangement of the input, including all repeated values. We prove $P(k)$ for $0 \leq k \leq n - 1$ by induction on k .

Base case ($k = 0$). Before the loop begins, the first zero positions form an empty prefix. The first two conditions are vacuously true: there are no prefix entries that could violate either condition. The array is still the original input, so the third condition also holds.

Why the inner loop finds a minimum. For a fixed outer index i , initially $m = i$, so $A[m]$ is a minimum among the one entry examined so far. When the next position j is examined, the algorithm replaces m by j if $A[j] < A[m]$ and otherwise retains m . In either case, $A[m]$ is a minimum among all entries examined through position j . Induction over these comparisons shows that, when the inner loop finishes, $A[m]$ is a minimum of $A[i], \dots, A[n]$. No array entries change during this search.

Question 7 continued

Inductive step. Suppose $0 \leq k < n - 1$ and $P(k)$ holds. The next outer-loop iteration has $i = k + 1$. By the minimum-search argument, the chosen value $t = A[m]$ is a minimum of the remaining entries in positions $k + 1, \dots, n$.

Both positions involved in the swap lie in this remaining part, so the first k entries are unchanged. The swap puts t into position $k + 1$. We verify each condition of $P(k + 1)$:

1. **The enlarged prefix is sorted.** By $P(k)$, the first k entries are already sorted and each is at most t , since t came from the remaining part. Appending t therefore gives a sorted prefix of length $k + 1$. When $k = 0$, the new prefix has one entry and is sorted automatically.
2. **The prefix is at most the remaining entries.** Each of the old prefix entries is at most every value that remains, by $P(k)$. The new prefix entry t is also at most every remaining value because it was a minimum of the entire old remaining part. Thus every entry in the enlarged prefix is at most every entry in positions $k + 2, \dots, n$.
3. **All occurrences are preserved.** The search changes only indices, and the swap only exchanges two entries. Neither operation changes the number of occurrences of any value.

Thus $P(k + 1)$ holds. Induction proves the claim throughout the loop.

(c) Termination and correctness.

The outer loop has $n - 1$ iterations. Each inner loop scans finitely many positions, and each comparison and swap finishes. Hence the algorithm terminates.

For $n \geq 2$, the final invariant is $P(n - 1)$. The first $n - 1$ entries are sorted, and each is at most the last entry. Therefore all n entries are in nondecreasing order. The occurrence condition ensures that this sorted array contains exactly the original input entries.

For $n = 1$, no iterations are needed: the original one-entry array is already sorted and is returned unchanged. The proof allows repeated values because it uses non-strict inequalities in the invariant and preserves the number of occurrences of every value. A minimum need not be unique.